

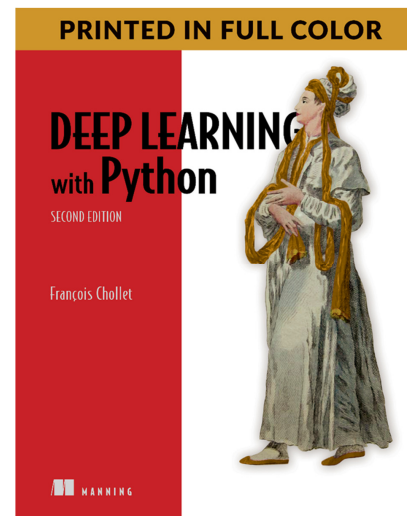
CS 59300

Application of Deep Learning Advanced DL for Computer Vision

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Reference



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Outline

- Three Essential Computer Vision Tasks
- An Image Segmentation Example
- Modern Convnet Architecture Patterns

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Three Essential Computer Vision Tasks

- Image classification
 - Assign one or more labels to an image.
- Image segmentation
 - “Segment” or “partition” an image into different areas, with each area usually representing a category.
- Object detection
 - Draw rectangles (called bounding boxes) around objects of interest in an image, and associate each rectangle with a class.

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Image Classification

Single-label multi-class classification



- ☒ Biking
- ☐ Running
- ☐ Swimming

Multi-label classification



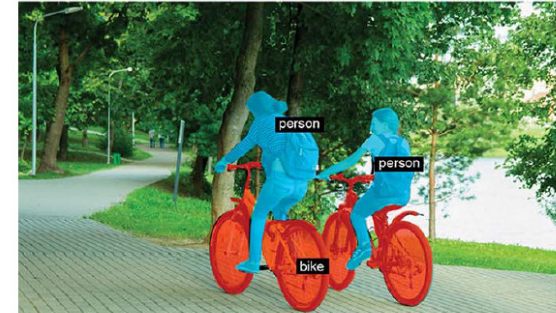
- ☒ Bike
- ☒ Person
- ☐ Boat
- ☒ Tree
- ☐ Car
- ☐ House

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Image Segmentation

- Example: Zoom or Google Meet displays a custom background behind you in a video call.

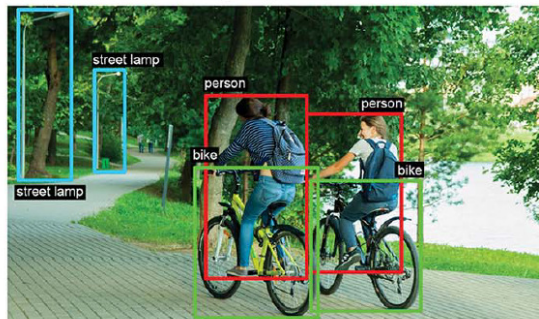
Image segmentation



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Object Detection

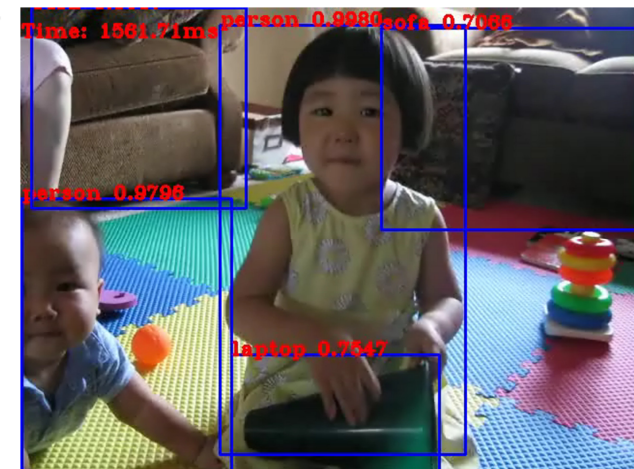
Object detection



- Example: Yolo

- <https://pjreddie.com/darknet/yolo/>
- <https://github.com/pjreddie/darknet>

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- **An Image Segmentation Example**
- Modern Convnet Architecture Patterns

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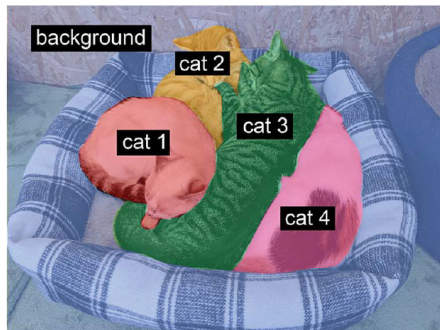
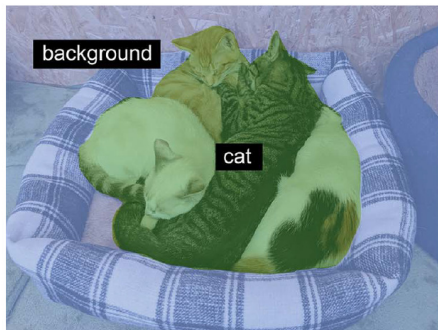
An Image Segmentation Example

- Image segmentation
 - Assign a class to each pixel in an image
 - Segmenting the image into different zones
 - E.g., “background” and “foreground,” or “road,” “car,” and “sidewalk”.
- Applications
 - Image and video editing
 - Autonomous driving
 - Robotics
 - Medical imaging
 - ...

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Semantic Segmentation Vs. Instance Segmentation

- **Semantic segmentation**: each pixel is independently classified into a semantic category, like “cat.”
- **Instance segmentation**: seeks not only to classify image pixels by category, but also to parse out **individual** object instances.



An Example

■ Oxford-IIIT Pets dataset

- <https://www.robots.ox.ac.uk/~vgg/data/pets/>
- 7,390 pictures of various breeds of cats and dogs
- Foreground-background segmentation masks for each picture

■ Segmentation mask

- The image-segmentation equivalent of a label
- An image the same size as the input image, with a single color channel where each integer value corresponds to the class of the corresponding pixel in the input image.

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Build the CNN Model

- The purpose of this first half is to **encode** the images into smaller feature maps, where each spatial location (or pixel) contains information about a large spatial chunk of the original image.
 - Compression
- **Downsample** by adding **strides** to every other convolution layer
 - Not use MaxPooling2D layers
 - Care a lot about the **spatial location** of the information in the image
- The second half of the model is a stack of **Conv2DTranspose** layers.

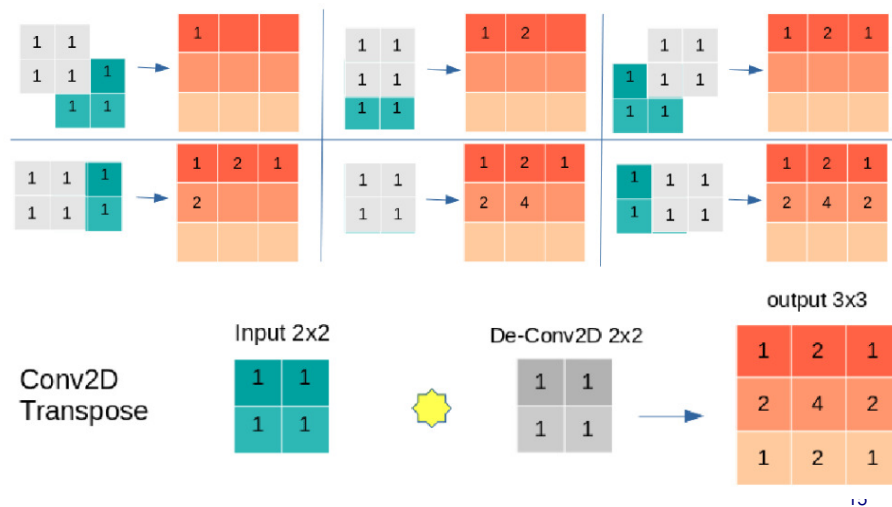
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Conv2DTranspose Layers

- Transposed convolution layer
- Sometimes called **Deconvolution**
- Inverse of Conv2D operation
 - Upsample
- Example:
 - <https://stackoverflow.com/questions/68976745/in-keras-what-is-the-difference-between-conv2dtranspose-and-conv2d>

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Conv2DTranspose Layers



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Outline

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- An Image Segmentation Example
- **Modern Convnet Architecture Patterns**

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Model Architecture

- The **sum** of the **choices** that went into creating it
 - Which layers to use
 - How to configure them
 - In what arrangement to connect them
- Define the **hypothesis space** of the model
- Like feature engineering, a good hypothesis space encodes **prior knowledge** that you have about the problem at hand and its solution.
 - Convolution layers – translation invariant

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Model Architecture

- A good model architecture
 - **Reduces** the size of the search space
 - Makes it **easier** to converge to a good point of the search space
- Model architecture is more an **art** than a science
 - Apply **intuition**
 - Maybe, no one can give you a clear **explanation** of what works and what doesn't
 - Pattern-match from extensive practical **experience**

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Best Practices for CNN Architecture

- Residual connections
- Batch normalization
- Separable convolutions

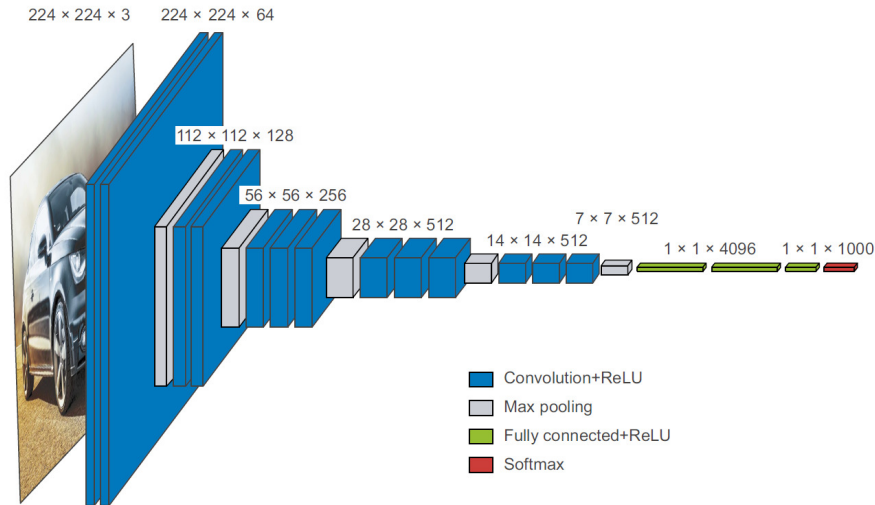
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Modularity-Hierarchy-Reuse (MHR) Formula

- Structure your amorphous soup of complexity into **modules**, organize the modules into a **hierarchy**, and start **reusing** the same modules in multiple places as appropriate
 - “Reuse” is another word for **abstraction** in this context
 - Formula for making a complex system **simpler**
 - Do not Repeat Yourself (**DRY**)
 - DL model architecture is **primarily** about making clever use of MHR

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VGG16 Architecture



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Residual Connections

Why residual connections?

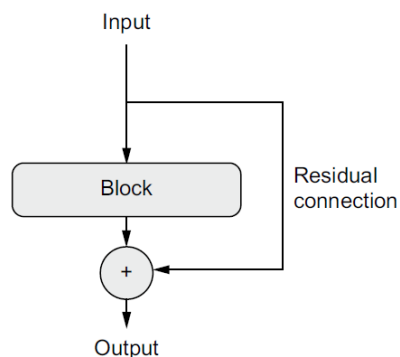
- In general, a **deep** stack of **narrow** layers performs better than a **shallow** stack of **large** layers.
- **Backpropagation** for a chain of functions.
- The problem of **vanishing gradients**.

Solution

- Force each function in the chain to be **nondestructive**.
- **Retain** a noiseless version of the information contained in the previous input.

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A Residual Connection Around a Processing Block



Pseudocode

```
X = ...
residual = x
x = block(x)
x = add([x, residual])
```

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Residual Block Where the Number of Filters Changes

```
from tensorflow import keras
from tensorflow.keras import layers

inputs = keras.Input(shape=(32, 32, 3))
x = layers.Conv2D(32, 3, activation="relu")(inputs)
residual = x
x = layers.Conv2D(64, 3, activation="relu", padding="same")(x)
residual = layers.Conv2D(64, 1)(residual)
x = layers.add([x, residual])
```

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Case Where the Target Block Includes a Max Pooling Layer

```
inputs = keras.Input(shape=(32, 32, 3))
x = layers.Conv2D(32, 3, activation="relu")(inputs)
residual = x
x = layers.Conv2D(64, 3, activation="relu", padding="same")(x)
x = layers.MaxPooling2D(2, padding="same")(x)
residual = layers.Conv2D(64, 1, strides=2)(residual)
x = layers.add([x, residual])
```

■ Another Example

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Batch Normalization

■ What is normalization?

- A broad category of methods that seek to make **different** samples seen by a machine learning model more **similar** to each other.

□ Standardization

```
normalized_data = (data - np.mean(data, axis=...)) / np.std(data, axis=...)
```

■ Why batch normalization?

- Help gradient propagation (like residual connections)

■ Why it works?

- Do **NOT** know

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Two Ways to Use Batch Normalization

■ The activation comes first

```
x = layers.Conv2D(32, 3, activation="relu")(x)
x = layers.BatchNormalization()(x)
```

■ The activation comes last

```
x = layers.Conv2D(32, 3, use_bias=False)(x)
x = layers.BatchNormalization()(x)
x = layers.Activation("relu")(x) ←
```

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Depthwise Separable Convolutions

■ Why depthwise separable convolutions?

- Make model **smaller** (fewer trainable weights) and **leaner** (fewer floating-point operations)
- Perform a few percentage points **better**

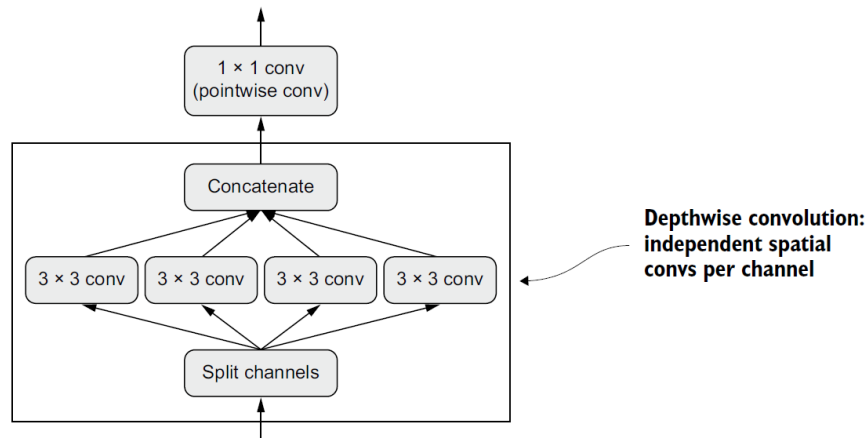
■ SeparableConv2D in Keras

- Performs a spatial convolution on each channel of its input, **independently**, before **mixing** output channels via a pointwise convolution (a 1×1 convolution)

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Depthwise Separable Convolutions



Assumption of Depthwise Separable Convolutions

- Spatial locations in intermediate activations are highly correlated, but **different channels** are highly **independent**.
- This assumption is generally **true** for the image representations learned by deep neural networks.
 - Help train small models from scratch on limited data.
- The assumption is not true for raw data inputs
 - E.g., RGB channels

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Some CNN Architecture Principles

- Organized into repeated **blocks** of layers
 - multiple convolution layers and a max pooling layer
- Number of **filters** should increase as the size of the spatial feature maps decreases
- **Deep** and **narrow** is better than broad and shallow
- Use residual connections
- Use batch normalization
- Try to replace Conv2D layers with SeparableConv2D layers

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Putting it together:
A mini Xception-like
model

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Summary

- Three essential computer vision tasks
 - Image classification / image segmentation / object detection
- An Image Segmentation Example
 - Downsample by adding strides to every other convolution layer
 - Apply Conv2DTranspose Layers
- Three best practices for modern CNN architecture
 - Residual connections / batch normalization / depthwise separable convolutions

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Q & A

Have fun!!!